

Here $C^n(A+B;G)$ is the dual of the subgroup $C_n(A+B) \subset C_n(X)$ consisting of sums of singular n -simplices lying in A or in B . The inclusion $C_n(A+B) \subset C_n(X)$ is a chain homotopy equivalence by Proposition 2.21, so the dual restriction map $C^n(X;G) \rightarrow C^n(A+B;G)$ is also a chain homotopy equivalence, hence induces an isomorphism on cohomology as shown in the discussion of excision a couple pages back. The map ψ has coordinates the two restrictions to A and B , and φ takes the difference of the restrictions to $A \cap B$, so it is obvious that φ is onto with kernel the image of ψ .

There is a relative Mayer-Vietoris sequence

$$\cdots \rightarrow H^n(X, Y; G) \rightarrow H^n(A, C; G) \oplus H^n(B, D; G) \rightarrow H^n(A \cap B, C \cap D; G) \rightarrow \cdots$$

for a pair $(X, Y) = (A \cup B, C \cup D)$ with $C \subset A$ and $D \subset B$ such that X is the union of the interiors of A and B while Y is the union of the interiors of C and D . To derive this, consider first the map of short exact sequences of cochain complexes

$$\begin{array}{ccccccc} 0 & \longrightarrow & C^n(X, Y; G) & \longrightarrow & C^n(X; G) & \longrightarrow & C^n(Y; G) \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & C^n(A+B, C+D; G) & \longrightarrow & C^n(A+B; G) & \longrightarrow & C^n(C+D; G) \longrightarrow 0 \end{array}$$

Here $C^n(A+B, C+D; G)$ is defined as the kernel of $C^n(A+B; G) \rightarrow C^n(C+D; G)$, the restriction map, so the second sequence is exact. The vertical maps are restrictions. The second and third of these induce isomorphisms on cohomology, as we have seen, so by the five-lemma the first vertical map also induces isomorphisms on cohomology. The relative Mayer-Vietoris sequence is then the long exact sequence associated to the short exact sequence of cochain complexes

$$0 \rightarrow C^n(A+B, C+D; G) \xrightarrow{\psi} C^n(A, C; G) \oplus C^n(B, D; G) \xrightarrow{\varphi} C^n(A \cap B, C \cap D; G) \rightarrow 0$$

This is exact since it is the dual of the short exact sequence

$$0 \rightarrow C_n(A \cap B, C \cap D) \rightarrow C_n(A, C) \oplus C_n(B, D) \rightarrow C_n(A+B, C+D) \rightarrow 0$$

constructed in §2.2, which splits since $C_n(A+B, C+D)$ is free with basis the singular n -simplices in A or in B that do not lie in C or in D .

Exercises

1. Show that $\text{Ext}(H, G)$ is a contravariant functor of H for fixed G , and a covariant functor of G for fixed H .
2. Show that the maps $G \xrightarrow{n} G$ and $H \xrightarrow{n} H$ multiplying each element by the integer n induce multiplication by n in $\text{Ext}(H, G)$.
3. Regarding $\mathbb{Z}_2$ as a module over the ring $\mathbb{Z}_4$, construct a resolution of $\mathbb{Z}_2$ by free modules over $\mathbb{Z}_4$ and use this to show that $\text{Ext}_{\mathbb{Z}_4}^n(\mathbb{Z}_2, \mathbb{Z}_2)$ is nonzero for all n .

4. What happens if one defines homology groups $h_n(X; G)$ as the homology groups of the chain complex $\cdots \rightarrow \text{Hom}(G, C_n(X)) \rightarrow \text{Hom}(G, C_{n-1}(X)) \rightarrow \cdots$? More specifically, what are the groups $h_n(X; G)$ when $G = \mathbb{Z}$, $\mathbb{Z}_m$, and $\mathbb{Q}$?

5. Regarding a cochain $\varphi \in C^1(X; G)$ as a function from paths in X to G , show that if φ is a cocycle, then

- (a) $\varphi(f \cdot g) = \varphi(f) + \varphi(g)$,
- (b) φ takes the value 0 on constant paths,
- (c) $\varphi(f) = \varphi(g)$ if $f \simeq g$,
- (d) φ is a coboundary iff $\varphi(f)$ depends only on the endpoints of f , for all f .

[In particular, (a) and (c) give a map $H^1(X; G) \rightarrow \text{Hom}(\pi_1(X), G)$, which the universal coefficient theorem says is an isomorphism if X is path-connected.]

6. (a) Directly from the definitions, compute the simplicial cohomology groups of $S^1 \times S^1$ with $\mathbb{Z}$ and $\mathbb{Z}_2$ coefficients, using the Δ -complex structure given in §2.1.

(b) Do the same for $\mathbb{R}P^2$ and the Klein bottle.

7. Show that the functors $h^n(X) = \text{Hom}(H_n(X), \mathbb{Z})$ do not define a cohomology theory on the category of CW complexes.

8. Many basic homology arguments work just as well for cohomology even though maps go in the opposite direction. Verify this in the following cases:

- (a) Compute $H^i(S^n; G)$ by induction on n in two ways: using the long exact sequence of a pair, and using the Mayer-Vietoris sequence.
- (b) Show that if A is a closed subspace of X that is a deformation retract of some neighborhood, then the quotient map $X \rightarrow X/A$ induces isomorphisms $H^n(X, A; G) \approx \tilde{H}^n(X/A; G)$ for all n .
- (c) Show that if A is a retract of X then $H^n(X; G) \approx H^n(A; G) \oplus H^n(X, A; G)$.

9. Show that if $f: S^n \rightarrow S^n$ has degree d then $f^*: H^n(S^n; G) \rightarrow H^n(S^n; G)$ is multiplication by d .

10. For the lens space $L_m(\ell_1, \dots, \ell_n)$ defined in Example 2.43, compute the cohomology groups using the cellular cochain complex and taking coefficients in $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{Z}_m$, and $\mathbb{Z}_p$ for p prime. Verify that the answers agree with those given by the universal coefficient theorem.

11. Let X be a Moore space $M(\mathbb{Z}_m, n)$ obtained from S^n by attaching a cell e^{n+1} by a map of degree m .

(a) Show that the quotient map $X \rightarrow X/S^n = S^{n+1}$ induces the trivial map on $\tilde{H}_i(-; \mathbb{Z})$ for all i , but not on $H^{n+1}(-; \mathbb{Z})$. Deduce that the splitting in the universal coefficient theorem for cohomology cannot be natural.

(b) Show that the inclusion $S^n \hookrightarrow X$ induces the trivial map on $\tilde{H}^i(-; \mathbb{Z})$ for all i , but not on $H_n(-; \mathbb{Z})$.

12. Show $H^k(X, X^n; G) = 0$ if X is a CW complex and $k \leq n$, by using the cohomology version of the second proof of the corresponding result for homology in Lemma 2.34.

13. Let $\langle X, Y \rangle$ denote the set of basepoint-preserving homotopy classes of basepoint-preserving maps $X \rightarrow Y$. Using Proposition 1B.9, show that if X is a connected CW complex and G is an abelian group, then the map $\langle X, K(G, 1) \rangle \rightarrow H^1(X; G)$ sending a map $f: X \rightarrow K(G, 1)$ to the induced homomorphism $f_*: H_1(X) \rightarrow H_1(K(G, 1)) \approx G$ is a bijection, where we identify $H^1(X; G)$ with $\text{Hom}(H_1(X), G)$ via the universal coefficient theorem.

3.2 Cup Product

In the introduction to this chapter we sketched a definition of cup product in terms of another product called cross product. However, to define the cross product from scratch takes some work, so we will proceed in the opposite order, first giving an elementary definition of cup product by an explicit formula with simplices, then afterwards defining cross product in terms of cup product. The other approach of defining cup product via cross product is explained at the end of §3.B.

To define the cup product we consider cohomology with coefficients in a ring R , the most common choices being $\mathbb{Z}$, $\mathbb{Z}_n$, and $\mathbb{Q}$. For cochains $\varphi \in C^k(X; R)$ and $\psi \in C^\ell(X; R)$, the **cup product** $\varphi \smile \psi \in C^{k+\ell}(X; R)$ is the cochain whose value on a singular simplex $\sigma: \Delta^{k+\ell} \rightarrow X$ is given by the formula

$$(\varphi \smile \psi)(\sigma) = \varphi(\sigma| [v_0, \dots, v_k]) \psi(\sigma| [v_k, \dots, v_{k+\ell}])$$

where the right-hand side is the product in R . To see that this cup product of cochains induces a cup product of cohomology classes we need a formula relating it to the coboundary map:

Lemma 3.6. $\delta(\varphi \smile \psi) = \delta\varphi \smile \psi + (-1)^k \varphi \smile \delta\psi$ for $\varphi \in C^k(X; R)$ and $\psi \in C^\ell(X; R)$.

Proof: For $\sigma: \Delta^{k+\ell+1} \rightarrow X$ we have

$$\begin{aligned} (\delta\varphi \smile \psi)(\sigma) &= \sum_{i=0}^{k+1} (-1)^i \varphi(\sigma| [v_0, \dots, \hat{v}_i, \dots, v_{k+1}]) \psi(\sigma| [v_{k+1}, \dots, v_{k+\ell+1}]) \\ (-1)^k (\varphi \smile \delta\psi)(\sigma) &= \sum_{i=k}^{k+\ell+1} (-1)^i \varphi(\sigma| [v_0, \dots, v_k]) \psi(\sigma| [v_k, \dots, \hat{v}_i, \dots, v_{k+\ell+1}]) \end{aligned}$$

When we add these two expressions, the last term of the first sum cancels the first term of the second sum, and the remaining terms are exactly $\delta(\varphi \smile \psi)(\sigma) = (\varphi \smile \psi)(\partial\sigma)$ since $\partial\sigma = \sum_{i=0}^{k+\ell+1} (-1)^i \sigma| [v_0, \dots, \hat{v}_i, \dots, v_{k+\ell+1}]$. $\square$